

ADCS Homework 4

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I. ACTUATOR SPECIFICATIONS

The ISS uses two classes of ADCS actuators: four large double-gimbal control moment gyros (CMGs) on the Z1 truss for primary non-propulsive attitude control, and the Zvezda service-module RCS cluster for momentum desaturation and reboost. Public information on the exact mounting geometry of the CMGs is incomplete, so we model the array as the standard symmetric 4-CMG pyramid used throughout the literature [1], [2], and approximate each CMG as a single-gimbal unit at the cluster level (which is how the Bedrossian et al. analyses treat them for momentum-envelope and steering-law design).

Both actuator groups are instantiated through the `mujoco_orbit` public API as `ControlMomentGyroSpec` and `ThrusterSpec`, compiled into `MjoModel` against `iss_model.xml`. All specifications, geometry, and Jacobians below are produced by `ISS/hw4/actuator_specs.py`.

The body frame convention is LVLH-aligned at rest: $+X$ along-track (roll), $+Y$ orbit-normal (pitch), $+Z$ nadir (yaw).

A. CMG Cluster

a) Device-level specifications.: Each CMG is a Honeywell M1000-class unit with the parameters summarized in Table I [1], [3]. The published rotor-momentum magnitude is $h = 4760 \text{ N m s}$ and the peak output torque is $\tau_{\max} = 258 \text{ N m}$; the maximum gimbal rate follows from these as $\dot{\theta}_{\max} = \tau_{\max}/h \approx 0.054 \text{ rad/s} \approx 3.1 \text{ deg/s}$.

TABLE I
PER-CMG SPECIFICATIONS (HONEYWELL M1000).

Quantity	Value
Rotor angular momentum h	4760 N m s
Peak output torque $\tau_{\max}$	258 N m
Max gimbal rate $\dot{\theta}_{\max}$	0.054 rad/s (3.11 deg/s)
Gimbal angle range	full rotation (wrap-around)

b) Pyramid array geometry.: Four CMGs are mounted on the Z1 truss with gimbal axes arranged in the standard symmetric pyramid with skew angle $\beta = \cos^{-1}(1/\sqrt{3}) = 54.7356^\circ$, which maximizes the inscribed momentum sphere [1]. We orient the pyramid apex along $+Z_{\text{body}}$. With azimuths $\phi_i \in \{0, \pi/2, \pi, 3\pi/2\}$, the gimbal axes $\hat{g}_i$, the spin axes at neutral gimbal $\hat{s}_i(0)$, and the output (torque) axes $\hat{t}_i(0) = \hat{g}_i \times \hat{s}_i(0)$ are listed in Table II. The neutral spin axes are chosen so that the net array momentum vanishes at $\theta = 0$:

$$\mathbf{H}(\mathbf{0}) = \sum_{i=1}^4 h \hat{s}_i(0) = \mathbf{0}. \quad (1)$$

TABLE II
CMG GEOMETRY IN THE BODY FRAME ($\beta = 54.7356^\circ$, APEX $\parallel +Z_{\text{BODY}}$).

CMG	$\hat{g}_i$	$\hat{s}_i(0)$	$\hat{t}_i(0)$
1 ($\phi = 0$)	(+0.816, 0, +0.577)	(0, +1, 0)	(-0.577, 0, +0.816)
2 ($\phi = \pi/2$)	(0, +0.816, +0.577)	(-1, 0, 0)	(0, -0.577, +0.816)
3 ($\phi = \pi$)	(-0.816, 0, +0.577)	(0, -1, 0)	(+0.577, 0, +0.816)
4 ($\phi = 3\pi/2$)	(0, -0.816, +0.577)	(+1, 0, 0)	(0, +0.577, +0.816)

c) Cluster momentum and Jacobian.: For an SGCMG at gimbal angle θ_i , the spin axis in the body frame is

$$\hat{s}_i(\theta_i) = \cos \theta_i \hat{s}_i(0) + \sin \theta_i \hat{t}_i(0). \quad (2)$$

The cluster angular-momentum map and the body-frame torque Jacobian with respect to commanded gimbal rates $\dot{\theta} \in \mathbb{R}^4$ are therefore

$$\mathbf{H}(\theta) = \sum_{i=1}^4 h \hat{s}_i(\theta_i), \quad (3)$$

$$\mathbf{A}(\theta) = \frac{\partial \tau_{\text{body}}}{\partial \dot{\theta}} = -h [\hat{t}_1(\theta_1) \mid \hat{t}_2(\theta_2) \mid \hat{t}_3(\theta_3) \mid \hat{t}_4(\theta_4)], \quad (4)$$

where $\hat{t}_i(\theta_i) = \hat{g}_i \times \hat{s}_i(\theta_i)$. The negative sign in (4) comes from Newton's third law: the body torque is the reaction to $d\mathbf{h}_i^{\text{body}}/dt = h \dot{\theta}_i \hat{t}_i$.

Evaluated at the neutral gimbal state $\theta = \mathbf{0}$,

$$\mathbf{A}(\mathbf{0}) = \begin{bmatrix} +2748 & 0 & -2748 & 0 \\ 0 & +2748 & 0 & -2748 \\ -3887 & -3887 & -3887 & -3887 \end{bmatrix} \text{ N m/rad/s}. \quad (5)$$

Rows 1 and 2 have the sparsity pattern $\pm h \cos \beta \cdot \hat{s}_i^{xy}$ (only opposing-azimuth CMGs contribute to roll/pitch torque at $\theta = \mathbf{0}$), and the row-3 entries are all $-h \sin \beta = -3887 \text{ N m}$ per rad/s, so all four CMGs contribute constructively to yaw torque in the zero-gimbal configuration. This is a well-known feature of the pyramid array and is why the "null motion" direction at $\theta = \mathbf{0}$ ($\dot{\theta} = (1, 1, 1, 1)$) produces a pure $-Z$ torque of magnitude $4h \sin \beta \cdot \dot{\theta}_{\max} \approx 842 \text{ N m}$.

d) Momentum envelope.: Sampling $\theta \in [-\pi, \pi]^4$ uniformly and plotting $\mathbf{H}(\theta)$ gives the solid region shown in Fig. 1. Its maximum extent is $4h = 19040 \text{ N m s}$ (when all spin axes align), and the largest inscribed sphere has radius $2h \cos(\beta/2) \approx 7558 \text{ N m s}$.

e) Output-torque set.: At $\theta = \mathbf{0}$, the image of the unit $\dot{\theta}$ -ball under $\mathbf{A}(\mathbf{0})$, scaled by $\dot{\theta}_{\max}$, is the rank-3 ellipsoid of body torques simultaneously achievable without saturating any gimbal (Fig. 2). The ellipsoid is stretched along $\pm Z$ because row 3 of (5) is coherent across all four CMGs.

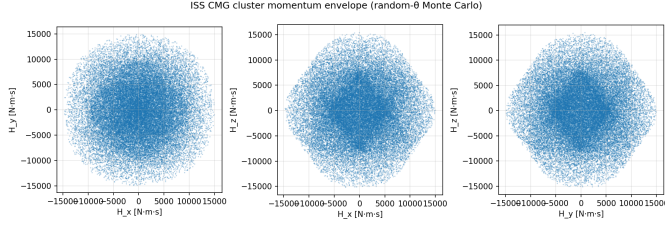


Fig. 1. Cluster angular-momentum envelope of the 4-CMG pyramid (40 000 random gimbal configurations). Projections on the three body-frame coordinate planes. The envelope is convex in each projection; internal singular surfaces exist within the solid region (not shown).

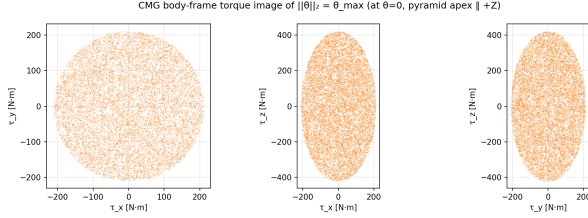


Fig. 2. Image of $\|\dot{\theta}\|_2 = \dot{\theta}_{\max}$ under $A(0)$ — the body-frame torques reachable without exceeding any gimbal-rate limit at the zero-gimbal configuration.

B. Zvezda RCS Thrusters

The Russian Zvezda service module provides the ISS's only on-station propulsive ADCS authority. Its aft propulsion assembly (PAO) houses ~ 32 small DPO attitude thrusters (130 N class) [4] and 2 large DKD reboost engines (3070 N class). For this analysis we use a simplified symmetric layout of 8 attitude thrusters on the aft rings plus the 2 aft-facing reboost engines (Table III). Positions are given with respect to the ISS center of mass, which is approximately 30 m forward of Zvezda's aft bulkhead.

a) Wrench Jacobian.: Each thruster produces a body-frame force $\hat{d}_i F_i$ and a torque $\mathbf{r}_i \times \hat{d}_i F_i$ about the CM, so the full $6 \times N$ wrench Jacobian is

$$B = \begin{bmatrix} \hat{d}_1 & \cdots & \hat{d}_N \\ \mathbf{r}_1 \times \hat{d}_1 & \cdots & \mathbf{r}_N \times \hat{d}_N \end{bmatrix}, \quad \begin{bmatrix} \mathbf{F} \\ \boldsymbol{\tau} \end{bmatrix}_{\text{body}} = B \mathbf{u}, \quad (6)$$

with $\mathbf{u} \in [0, F_{\max,i}]^N$. The numerical evaluation gives

$$B = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & +1 & +1 \\ +1 & -1 & +1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & +1 & +1 & -1 & -1 & 0 & 0 \\ -3 & +3 & +3 & -3 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & +29 & +31 & -29 & -31 & 0 & 0 \\ -30 & +30 & -30 & +30 & 0 & 0 & 0 & 0 & -0.8 & +0.8 \end{bmatrix}. \quad (7)$$

The top three rows give body-frame force per unit thrust and the bottom three give torque per unit thrust. Firing each pair of Y-thrusters in the same direction produces a pure roll or yaw couple; firing each Z-pair in the same direction produces a pure pitch couple; the two DKD engines provide pure along-track force (with a small residual yaw couple from their $\pm Y$ lever arms).

b) Maximum body wrenches.: Saturating the thrusters that contribute in a given axis direction gives the upper-bound pure-torque capabilities in Table IV. Roll authority is the weakest (limited to the ± 3 m ring radius), while yaw authority is the strongest (benefiting from the ~ 30 m lever arm along $-X$).

c) Use in the overall ADCS.: Although the thrusters have much higher instantaneous torque than the CMG cluster, they are propellant-limited and are not used for continuous attitude regulation. Their role is to perform CMG desaturation (momentum dumping) when secular disturbance torques accumulate to the cluster momentum limit, and to execute orbit reboost. Quantitative estimates of how often desaturation is required are addressed in Sec. II (HW4 Q2), which is still to be written.

II. ENVIRONMENTAL TORQUES

To be completed.

III. ATTITUDE REGULATION

To be completed.

IV. EIGEN-AXIS SLEW

To be completed.

REFERENCES

- [1] B. Wie, *Space Vehicle Dynamics and Control*, 2nd ed. AIAA Education Series, 2008. Chapter 7.
- [2] N. Bedrossian, S. Bhatt, M. Lebsack, and E. Wong, "International Space Station zero-propellant maneuver," *Journal of Guidance, Control, and Dynamics*, vol. 32, no. 4, pp. 1193–1201, 2009.
- [3] Honeywell Defense and Space, "M1000 Control Moment Gyroscope," product information sheet.
- [4] NASA, "International Space Station Propulsion System," ISS Familiarization Manual, ISS FAM C TD9702A, 1998.

TABLE III
SIMPLIFIED ZVEZDA RCS LAYOUT MODELED IN ACTUATOR_SPECS.PY.

Role	Label	Position [m]	Direction	$F_{\max}$ [N]
Attitude (ring, top/bot $\pm Y$)	ACS_+Y_top	$(-30, +3, +3)$	$(0, +1, 0)$	130
	ACS_-Y_top	$(-30, -3, +3)$	$(0, -1, 0)$	130
	ACS_+Y_bot	$(-30, +3, -3)$	$(0, +1, 0)$	130
	ACS_-Y_bot	$(-30, -3, -3)$	$(0, -1, 0)$	130
Attitude (ring, fwd/aft $\pm Z$)	ACS_+Z_fwd	$(-29, 0, +3)$	$(0, 0, +1)$	130
	ACS_+Z_aft	$(-31, 0, +3)$	$(0, 0, +1)$	130
	ACS_-Z_fwd	$(-29, 0, -3)$	$(0, 0, -1)$	130
	ACS_-Z_aft	$(-31, 0, -3)$	$(0, 0, -1)$	130
Reboost (DKD)	DKD_stbd	$(-31.5, +0.8, 0)$	$(+1, 0, 0)$	3070
	DKD_port	$(-31.5, -0.8, 0)$	$(+1, 0, 0)$	3070

TABLE IV
UPPER-BOUND PURE-TORQUE AND REBOOST CAPABILITIES FROM THE
SIMPLIFIED ZVEZDA LAYOUT.

Command	Max magnitude
Pure roll torque τ_x	780 N m
Pure pitch torque τ_y	7800 N m
Pure yaw torque τ_z	10 256 N m
Pure $+X$ force (reboost)	6140 N